Homework 2 | **March 1, 11.59 pm**
CS 09410 and CS 09510 | Spring 2026 | Prof. André Gomes
Topics: Chapter 2

Read the following instructions before submitting your work:

- Submitting work copied (in part or in whole) from other persons, websites, or generative AI tools such as ChatGPT is strictly forbidden.
- Submit your answers on Canvas as a single PDF file.
- Include a heading with your name and assignment number/date.
- Handwriting is allowed if writing is readable by others.

Problem 1: Summarize the main characteristics of TCP and UDP that should be taken into account when deciding which protocol to use for an Internet application.

Problem 2: Consider the network illustrated in Figure 1. On the left, the Internet service provider (ISP) provides Internet connectivity to approximately 5 million users located in the West coast of the US. On the right, a data center, located in the East coast, hosts the Web server for a popular newspaper. The link capacity between ISP and the data center averages 10 Gbps. The link capacity between each user and the ISP network, 100 Mbps.

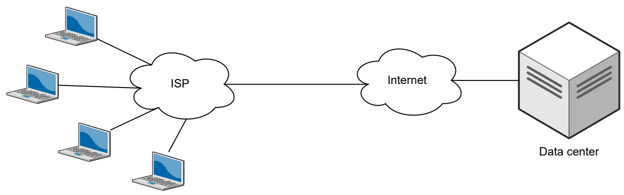

Figure 1: Example.

a) Users have frequently complained about long delays—or even the inability—to access the newspaper website. Explain why users may be experiencing long delays.

b) After some analysis, the ISP found that the demand for that newspaper website is particularly high in the mornings. Between 7 and 8 am, it handles $1,956$ requests per second on average, each request being a webpage of approximately 5 Mbits. It was also found that approximately 30% of those requests are to common web pages. Someone recommended that the ISP should install a Web cache server. Would you support that recommendation? Why or why not?

Problem 3: What is the HOL blocking issue in HTTP/1.1 and how does HTTP/2 attempt to solve it?

Problem 4: True or false:

a) A user requests a web page that consists of some text and three images. For this page, the client will send one request message and receive four response messages.

b) Two distinct web pages can be sent over the same persistent connection.

c) With non-persistent connections between browser and server, it is possible for a single TCP segment to carry two distinct HTTP request messages.

d) The `Date:` header in HTTP response message indicates when the object in the response was last modified.

e) HTTP response messages never have an empty message body.

Problem 5: Consider an HTTP client that wants to retrieve a web document at a given URL. The IP address of the HTTP server is initially unknown. What transport and application-layer protocols besides HTTP are needed in this scenario?

Problem 6: Consider a short, 10-meter link connecting a client and a server. The link supports communication at the 150 bits/sec rate in both directions.

Now consider the HTTP protocol. The client requests a 100 Kbits object that contains 10 referenced objects to be downloaded from the same server. For simplicity, assume that HTTP requests are negligible compared to the $\approx$ 100 Kbits HTTP responses.

Let the TCP connection setup time be $600/R$ sec[1], where R is the link bandwidth available for each connection in bits/sec. How long does it take to download the entire set of objects?

a) Assuming non-persistent HTTP with 1 TCP connection.

b) Now assume that objects can be downloaded in N parallel TCP connections, and that each connection gets $1/N$ of the link bandwidth.

c) Assume persistent HTTP with 1 TCP connection.

Does persistent HTTP always offer significant gains compared to non-persistent HTTP?

[1]For your information only: The underlying assumption is that each handshake message is 200 bits long, and it takes three messages (3-way handshake!) to setup a TCP connection.

Problem 7: In this problem, you will use the *dig* tool available on Unix and Linux hosts to explore the hierarchy of DNS servers. Before you start, explore the *man* page for *dig* (e.g., on the terminal, enter `man dig`). Particularly, familiarize yourself with the command:
dig +norecurse @<dns-server-ip> <name>.

Our goal is to obtain the server IP address for `rowan.edu` by iteratively querying root DNS server, top-level domain DNS server, and authoritative DNS server.

- Start with a root DNS server. You can find the list of root DNS servers here (click to open). Pick one, e.g., 198.41.0.4. Using *dig*, obtain the list of DNS servers in the delegation chain. Keep querying them until you obtain the IP address for `rowan.edu`. Report the chain of requests (screenshots!) you made and the obtained IP address for `rowan.edu`.

- Repeat the steps above for a website of your choice.